

POLYESTER

Definition

Polyester fibre was first made available to consumers (in US) in 1951 under the trade mark Decron by Du pont. It is defined as a manufactured fibre which the fibre forming substance is any long-chain synthetic polymer composed of at least 85% by weight of ester of a substituted aromatic carboxylic acid, including but not restricted to substituted terephthalic units and para substituted hydroxyl benzoate units.

Production of polyester fibers

Polyesters are the product of reactions between dihydric alcohol and dicarboxylic acid. They may be formed from substitute products as long as the end product is ester. In the manufacture of most polyester fibres, ethylene glycol is used as the dihydric alcohol. Terephthalic acid or Dimethyl terephthalate are used as the acid but dimethyl terephthalate has been used more frequently than terephthalic acid, because it is more readily available in pure form. Terephthalic acid or dimethyl terephthalate and ethylene glycol polymerise by condensation reaction to form the polyester polymer.

The two substances are polymerized by condensation at a high temperature. As the resulting polyester leaves the polymerisation vessel, it enters a cool area where the strands solidify. These strands which are about as thick as spaghetti, are cut into small pellets or chips dried and stored until needed for filament formation.

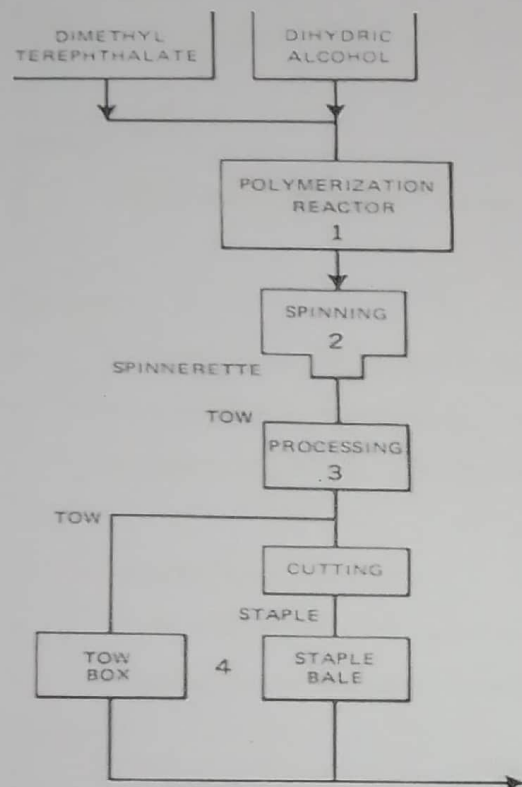
The actual formation is done in the same plant, or polyester pellets or chips may be transferred to other locations. The pellets or chips are melted and forced through fibre spinnerets. The filaments solidify in cooling air and are then taken up on containers resembling large spools. The filaments are then drawn and stretched to impact strength and control elongation.

The drawing process can be done partially as the filaments are extruded. These partially oriented yarns (POY) are used for making textured yarn. Bulked/textured/crimped

It should be noted that most manufacturers make polyester fibres by using a direct polymerisation to spin process and omit the pellet or chip stage. This is a continuous process from raw material to fibre. A large proportion of polyester fibres used today is in staple form i.e. cut to lengths of about 11/2" after the fibre has been crimped following the required

drawing. Staples fibres are frequently blended with cotton or other fibres and spun into yarns using a standard yarn manufacturing process.

FLOW DIAGRAM SHOWING PROCESSES IN POLYESTER MANUFACTURE



STARTING CHEMICALS: The production of KODE polyester fiber begins with dimethyl terephthalate and a special suitable dihydric alcohol.

1 POLYMERIZATION: The dihydric alcohol and dimethyl terephthalate are placed in a polymerization reactor where, under the proper conditions, they combine to form long chain-like molecules called **polymer**.

2 SPINNING:

The liquid polymer is forced through microscopic holes in a device similar to a miniature shower head, called a **spinnerette**.

The thin strands formed by the spinnerette are cooled and gathered into bundles of continuous fiber called **tow**, which looks like untwisted rope.

3 PROCESSING:

The tow is drawn and heat treated to make it resistant to shrinking or to stretching. Lubricants are added to aid in spinning yarns.

The tow may also be crimped like a "permanent wave" to add bulk and texture to the fiber.

The tow may be cut into short lengths called **staple** fiber or it may be left uncut, in continuous tow form ... depending upon the customer's (mill's) requirements.

4 PACKING:

Staple fiber is compressed into bales.

Continuous-form tow is packed into boxes.

SHIPPING: These forms of KODEL polyester fiber are then shipped to textile mills where the fibers are spun into yarns and woven into fabrics.

Tennessee Eastman Company produces the KODEL polyester fiber. It does not weave fabrics or make garments or home furnishings.

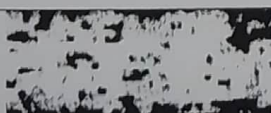
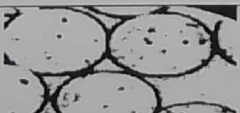
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Fibre Properties

Microscopic Appearance

Longitudinal view exhibits smooth surface and a rod-like appearance.

Cross-sectional view is round. Dacron is trilobol.

Longitudinal view	Cross-sectional view
	

Physical Properties

Shape and Appearance

- Polyester are generally round and uniform can be of any length or diameter.
- Fibre is partially transparent or white or slightly off-white

Lusture

- Dull or semi-dull appearance due to addition of pigments

Strength

- Tenacity varies from 1.5g/d to 9.5 g/d
- No loss in strength when wet. Low tenacity fibres are reserved for apparel, while higher tenacity ones (5 and above) are for industrial applications.

Elastic Recovery and Elongation

- Varies with tenacity strong fibres have lower degree of elongation
- Elastic recovery is superior to cellulosic fibre but inferior to nylon

Resiliency

- Is excellent recovery from crease and wrinkle
- Heat setting stabilizes fibre and yarn

Density

- Polyesters are similar to wool and acetates on their densities, i.e. 1.3-1.32g/d

Moisture Absorption

- They have low moisture regain of only 0.4-0.6 at standard conditions.
Even at 95 - 100% relative humidity moisture control is only 0.6-0.8%. It requires special techniques in dyeing and finishing.

Dimensional Stability

- Polyester will not shrink or stretch if properly heat set. Polyester that has not been heat-set may shrink at elevated temperatures.

Thermal Properties

- Polyester fibres melt at temperatures from 238°C - 290°C , form a grey or tan coloured bead that is hard and non crushable.
- They burn and produce a dark smoke and aromatic smell.

Chemical Properties

Effect of Alkalis

Polyester has good resistance to weak alkalis at high temperatures and room temperatures has moderate resistance to strong alkalis at room temperatures and it is degraded at elevated temperatures.

Effect of Acids

Weak acids even hot have not effect on polyester unless it is exposed for several days. Polyester have good resistance to strong acids at room temperature. Prolonged exposure to boiling hydrochloric acid destroys the fibre and 96% sulphuric acid causes disintegration.

Effect of Organic Solvents

Polyester is resistant to organic solvents.

Effect of Sunlight, Age and Miscellaneous Factors

Has good resistance to sunlight when behind glass. Prolonged exposure to sunlight weakens fibre. Not affected by ageing, resists abrasions.

Biological Properties

Insects do not destroy polyesters.

Use and Care of Polyester

- Apparel - is limitless varieties
- Blends of wool, rayon, cotton, linen with polyester are popular for apparel.
- Industry such as laundry bags, calendar, sheeting, press covers, conveyor belts, fire hose, sail cloth, fish netting, ropes, protective clothing.

- Used for surgical implants
- Unless it has been well stabilized during heating setting, it may shrink during care.
- Care of 100% polyester fabrics is minimal

Summary

What is polyester textile fiber?

Polyester fibre is a widely used synthetic fiber. Basic raw materials for polyester fibre production are petroleum, coal, air & water. From those basic raw materials the Glycol monomer dihydric alcohol and Terephthalic acid are produced. These are polymerized into an autoclave at high temperature and polyester chips are produced, which are melt spun.

General Properties of Polyester Textile:

1. Length: Polyester fibre could be produced in filament form or staple form i.e. as we desire. Fibre length is controllable.
2. Fineness of polyester textile: Fineness of polyester fibre is also controllable.
3. Cross-sectional shape: Normal cross sectional shape is round but it is also made triangular, elliptical or pentagonal. Normally it is white but could be of any color if color is added during spinning.
4. Strength: Normally tenacity varies from 4.5 to 5.0 gram per denier for the polyester textile. In case of high tenacity fibre it may be up to 8.0 gram per denier.
5. Extensibility: Extension at break varies from 20% to 30 %. Good recovery from extension. Due to good extension, strength and functional property polyester is widely used as sewing thread in the garment industries.
6. Resiliency: Polyester textile shows good resiliency property. It does not crease easily and any undue crease can be recovered easily.
7. Dimensional Stability: Polyester fibre is dimensionally stable. It could be heat-set at around 200 degree C. heat set polyester fibre does not shrink or extend.
8. Moisture Regain: Very low moisture ranges from 0 – 0.4%. Due to low moisture regain polyester fiber feels warm, easily cleaned, soils less.
9. Action of bleaching agents: It is not damaged by the action of bleaching agents.
10. Action of acid and alkali: It is unaffected by the action of acid and alkali.
11. Action of organic solvent: Polyester textile is unaffected by organic solvent, hence polyester fibre could be dry-cleaned.

12. Action of Sunlight & Heat: Polyester is unaffected by normal sunlight but prolonged exposure to sunlight can reduce strength of polyester. Polyester fibre melts at 227 degree C to 242 Degree C. hence during ironing care should be taken to reduce the risk of damage. In contact to flame the polyester burns and melt when cold, the molten polymer forms hard beads.

13. Dye ability: Polyester fibre shows good dye ability at high temperature. Color fastness to washing and sunlight is also good. Disperse dyes are widely used for dyeing polyester fibre.

14. Biological Properties: Resistance to attack of mildew, moths and insects are good.

15. End use of Polyester textiles: Polyester fibre is widely used for apparel in both 100% form and blended form with cotton, nylon, wool, rayon, acetate, etc fibres. Polyester textiles also used for making thermal underwear.